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Modes in optical waveguides formed by diffusion

Esther M. Conwell

Xerox Rochester Corporate Research Center, Rochester, New York 14644
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For an optical guide with index of refraction having a small exponential variation superimposed on a constant background value, such as is found in guides made by diffusing Se into CdS, the wave equation may be solved exactly. The solutions and characteristic equations for determining the possible TE and TM modes are given and the properties of the modes are discussed.

There has been considerable interest recently in optical waveguides formed by diffusion, thus circumventing the difficulty of growing suitable thin single-crystal layers.[1,2] For wide guides, i.e., guides very many wavelengths in width, it is possible to estimate the number of modes and their properties by means of the WKB approximation,[3] familiar as a method of obtaining approximate solutions of the Schrodinger equation. The WKB approximation is not applicable, however, close to a turning point—i.e., a region where the behavior changes from oscillatory to exponential. Thus, in a narrow guide where only a few oscillations, at most, separate regions of exponential behavior, the WKB is not useful.

To estimate the number of modes in their narrow guides formed by diffusing Se into CdS crystals, Taylor *et al.*[1] solved the wave equation exactly under the assumption that the refractive index varies linearly over the observed diffusion depth from the value deduced for the CdSSe surface to the value in pure CdS. According to their experimental results and to extensive earlier work on this system,[4] Se concentration varies with distance below the surface z as $\exp(-|z|/d)$, d being the diffusion depth. The departure from the expected Gaussian dependence on $|z|$ is not particularly unusual and may be due to the diffusing species being in two forms or occupying two different sites not in local equilibrium. Since the refractive index n has been found to increase linearly with Se concentration,[5] we expect the actual variation below the surface to be given by

$$n(z) = n_0 + \Delta n \exp(-|z|/d), \tag{1}$$

n_0 being the index for CdS (taken here as the average of ordinary and extraordinary indices) and Δn being the observed change at the surface. With $\Delta n \ll n_0$, the dielectric constant ϵ may be approximated by

$$\epsilon(z) = \epsilon_0 + \Delta\epsilon \exp(-|z|/d), \tag{2}$$

where $\epsilon_0 = n_0^2$ and $\Delta\epsilon = 2n_0\Delta n$. We find that the wave equation can be solved exactly for $\epsilon(z)$ given by (2). The modes may then be described quantitatively with much less effort than required with the linear approximation. Use of the latter entails boundary conditions at both $z=0$ and $z=-d$, whereas here only $z=0$ need be considered since the crystal is, in effect, semi-infinite.

For TE modes propagating in the x direction (in a planar guide infinite in the y direction), the additional terms in the wave equation arising from gradϵ vanish.[3] When we assume a solution

$$E \equiv E_y = F(z)\exp[i(k_x x - \omega t)], \tag{3}$$

the wave equation becomes

$$\frac{d^2F}{dz^2} + \left[\epsilon(z)\frac{\omega^2}{c^2} - k_x^2\right]F = 0. \tag{4}$$

For $\epsilon(z)$ given by (2) a solution of this equation can be obtained in terms of Bessel functions,

$$F(z) = AJ_{2dp_0}[4\pi(\Delta\epsilon)^{1/2}(d/\lambda)\exp(-|z|/2d)], \tag{5}$$

where A is an arbitrary constant, $\lambda = 2\pi c/\omega$, and

$$p_i = (k_x^2 - \epsilon_i\omega^2/c^2)^{1/2}, \quad i=0. \tag{6}$$

The other part of the solution, involving J_{-2dp_0} or Bessel functions of the second kind, has been dropped because it is not finite for $z=-\infty$.

The values of k_x must be chosen to satisfy the boundary conditions at $z=0$. Taking the medium above $z=0$ to be homogeneous with dielectric constant ϵ_2, we assume the usual exponential solution

$$E_y = B\exp(-p_2 z)\exp[i(k_x x - \omega t)], \quad z>0 \tag{7}$$

where B is an arbitrary constant and p_2 is given by (6) with $i=2$. The value of k_x has been taken to be equal to that for $z<0$, as will be necessary to satisfy the boundary conditions. The condition that E_y be continuous at $z=0$ leads, according to (7), (5), and (3), to a linear relation between A and B. Determining H_x from E_y by Maxwell's equations and requiring that H_x be continuous at $z=0$ leads to a second linear relation between A and B. Eliminating A and B from this pair of relations, we obtain the characteristic equation specifying the possible k_x values for the TE modes:

$$\frac{J_{2dp_0-1}(\xi) - J_{2dp_0+1}(\xi)}{J_{2dp_0}(\xi)} = \frac{-p_2\lambda}{\pi(\Delta\epsilon)^{1/2}}, \tag{8}$$

where $\xi = 4\pi(\Delta\epsilon)^{1/2}d/\lambda$. To obtain (8) in this form, the derivative of J_{2dp_0} was expressed in terms of Bessel functions of orders $2dp_0 \pm 1$.

To represent guided modes, since ϵ_0 and ϵ_2 are both assumed positive, the solutions must be oscillatory for some range of z below $z=0$. This requires $k_x^2 < (\epsilon_0 + \Delta\epsilon) \times (\omega/c)^2$. Combining this with the lower limit on k_x obtained from the condition that p_0 be real, we obtain for the range of k_x

$$(\epsilon_0 + \Delta\epsilon)^{1/2} > k_x(c/\omega) > \epsilon_0^{1/2}. \tag{9}$$

This relation ensures also the reality of p_2 since we assume $\epsilon_0 > \epsilon_2$. For $\Delta\epsilon \ll \epsilon$, the range of k_x values is quite small. As a consequence of (9), p_2 must lie within the range

$$(\epsilon_0 - \epsilon_2 + \Delta\epsilon)^{1/2} > (p_2\lambda/2\pi) > (\epsilon_0 - \epsilon_2)^{1/2}. \tag{10}$$

From (9) we also deduce

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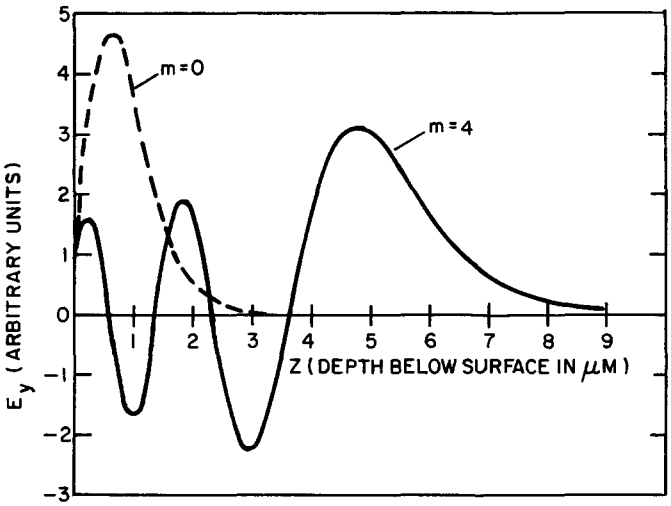

FIG. 1. Plots of $m=0$ and $m=4$ modes (unnormalized), calculated for a guide with an index of refraction varying according to $n=2.47[1+0.024\exp(-|z|/2.5)]$ for z in μm. $\lambda=0.6328$ μm.

$$4\pi(\Delta\epsilon)^{1/2}(d/\lambda) > 2dp_0 > 0. \tag{11}$$

For $\epsilon_2=1$ and $\epsilon_0=6.1$, as is the case for the guides of Ref. 1, the right-hand side of (8) varies little over the range (11) and, furthermore, for small $\Delta\epsilon$, is quite large, much larger than the J's. As a result, the characteristic values $k_x^{(i)}$ will appear at values of $2dp_0^{(i)}$, quite close to those for which $J_{2dp_0}[4\pi(\Delta\epsilon)^{1/2}d/\lambda]$ vanishes. The number of TE modes the guide will support is then equal to the number of $2dp_0$ values in the range (11) for which $J_{2dp_0}[4\pi(\Delta\epsilon)^{1/2}d/\lambda]$ vanishes.

Let us illustrate this for the case of Ref. 1, where $\Delta\epsilon=0.3$, $\lambda=0.633$ μm, and $d=2.5$ μm. The argument of the Bessel functions in (8) is then 27.0, and the range of $2dp_0$ is from 0 to 27.0. The zeroes of $J_\nu(27.0)$ lie at $\nu\simeq 21.5$, 17.4, 14.1, 11.2, 8.5, 6.1, 3.8, and 1.7,[6] all within the permitted range of $2dp_0$. Thus the guide will support eight TE modes. The estimate of Taylor *et al.*, based on the linear approximation described earlier, was three TE modes.[1] It is not surprising that the linear approximation underestimates the number of modes, since it arbitrarily cuts short the range of spatial variation of ϵ.

It is easily seen that the solution (5) gives the correct asymptotic behavior as z approaches $-\infty$. In that limit the argument of the Bessel function is very small. For small ξ, $J_\nu(\xi)$ varies as ξ^ν. From (5) we therefore find, for z approaching $-\infty$, $F\propto\exp(-p_0|z|)$, the expected behavior.

The transition from exponential to oscillatory behavior can be considered to take place at the turning or reflection point, $z_0^{(i)}$ for the ith mode, determinated by $(k_x^{(i)})^2=(\omega/c)^2\epsilon(z_0^{(i)})$. Thus the smaller the value of $k_x^{(i)}$, the larger the distance below the surface in which there is oscillatory behavior. Also, the smaller the value of $k_x^{(i)}$, the larger the number of oscillations. This is so because the number of nodes in the mode characterized by $k_x^{(i)}$, having order $\nu^{(i)}=2dp_0^{(i)}$, is the number of zeroes of $J_{\nu^{(i)}}(\xi)$ between $\xi=4\pi(\Delta\epsilon)^{1/2}d/\lambda$ and $\xi=0$. The smaller the value of $\nu^{(i)}$, the larger the number of nodes in this fixed range.[6] Another property of the oscillations is that their amplitude increases as z approaches $z_0^{(i)}$. In the present case this results from the property that the extreme values of $J_\nu(\xi)$ increase with decreasing ξ.[6] Landau and Lifshitz have shown that the increase in wave amplitude as the reflection point is reached is a typical property in the presence of a varying index of refraction.[3]

For the example we have been discussing, the modes may be numbered 0–7, in descending order of $\nu^{(i)}$ and $k_x^{(i)}$, according to the number of nodes in each. Modes $m=0$ and $m=4$ are shown in Fig. 1 and illustrate the various properties described above. Note that the highest peak in the $m=0$ mode is at $|z|<d/2$, while that in the $m=4$ mode is at $|z|\simeq 2d$.

For the TM modes, if we assume H_y in the same form as (3) for E_y, we may write the wave equation in the form[3]

$$\frac{\partial}{\partial z}\left(\frac{1}{\epsilon(z)}\frac{\partial H_y}{\partial z}\right)+\left(\frac{\omega^2}{c^2}-\frac{k_x^2}{\epsilon(z)}\right)H_y=0. \tag{12}$$

For $\Delta\epsilon$ small [i.e., $(\Delta\epsilon)^2\ll\epsilon$] and not varying too rapidly $[d\gg\lambda/2\pi(2\epsilon_0)^{1/2}]$, the terms involving derivatives of ϵ may be neglected. The substitution in (12) of $H_y=u(z)\epsilon^{1/2}(z)$ results in a differential equation for u identical with (4). Thus the solution for H_y inside the diffused medium differs from (5) only in having a multiplying factor $\epsilon^{1/2}(z)$. This leads to a characteristic equation for the TM modes, to terms linear in $\Delta\epsilon$:

$$\frac{J'_{2dp_0}(\xi)+[\lambda(\Delta\epsilon)^{1/2}/4\pi\epsilon_0 d]J_{2dp_0}(\xi)}{J_{2dp_0}(\xi)}=-\frac{\epsilon_0}{\epsilon_2}\frac{p_2\lambda}{2\pi(\Delta\epsilon)^{1/2}}\left(1+\frac{\Delta\epsilon}{\epsilon_0}\right), \tag{13}$$

where again $\xi=4\pi(\Delta\epsilon)^{1/2}d/\lambda$. This differs from the characteristic equation for the TE modes in having a small additional term in the numerator on each side and the factor ϵ_0/ϵ_2 on the right-hand side. The additional terms in the numerator will have little effect. The factor ϵ_0/ϵ_2, about 6 for the example we have been considering, will have the effect that for the TM modes the roots of the characteristic equation will lie even closer to the zeroes of $J_{2dp_0}[4\pi(\Delta\epsilon)^{1/2}d/\lambda]$. The possible number of TM modes will almost invariably be the same as the possible number of TE modes, and $H_y(z)$ will be quite similar to $E_y(z)$ since the factor $\epsilon^{1/2}(z)$ varies very little with z.

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